

Zihan Wang

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Objective

My long-term research vision is to advance **AI-enabled, experimentally grounded modeling and design methodologies for advanced manufacturing**, enabling the autonomous discovery, analysis, and scalable production of mechanical and material systems. I plan to pursue three interrelated research directions:

- **AI-based autonomous design and discovery of smart manufacturing:** developing AI-augmented autonomous design ecosystem that integrates high-throughput simulation, experiment, and adaptive learning within closed-loop, data-driven workflows.
- **Generative co-design of manufacturable mechanical and material systems:** enabling high-performance mechanical and material systems that can be fabricated reliably and at scale.
- **Uncertainty-aware design of mechanical and material systems for manufacturing:** modeling variability and heterogeneity to ensure robust and reliable performance.

Current Position

Postdoctoral Scholar	Sep 2024 – Present
Northwestern University, Evanston, IL, USA	
Advisor: Professor Wei Chen	
Research Direction: AI for materials discovery and design	

Education

Ph.D. in Mechanical Engineering	Aug 2019 – Aug 2024
University of Connecticut (UConn), CT, USA	
GPA: 4.061/4.0	
Advisor: Professor Hongyi Xu	
Dissertation: Design, Performance Evaluation and Uncertainty Analysis of Metamaterial Units	
B.S. in Materials Science and Engineering	Aug 2015 – May 2019
Huazhong University of Science and Technology (HUST), Wuhan, China	
GPA: 3.89/4.0 (Graduate with Honors)	
Major: Material Processing and Control Engineering	
B.A. in English Interpretation	Jan 2016 – May 2019
Huazhong University of Science and Technology (HUST), Wuhan, China	

Honors & Awards

Rising Star in Mechanical Engineering, MIT Workshop	2025
Held by MIT, around 30 candidates are selected	
Design Automation Dissertation Award, ASME Design Engineering Division	2025
Awarded annually to only one recent Ph.D. student who has made early notable contributions in the area of design automation.	
Pratt & Whitney Advanced Systems Engineering Travel Fellowship, UConn	2024
Awarded to 10 students per year in Systems Engineering at UConn, <i>Award amount: \$2000.</i>	
1st Place, ASME Computer & Information in Engineering Division Hackathon Event	2023
Awarded by ASME Computer and Information in Engineering Division, <i>Award amount: \$1400.</i>	

BPart Fellowship, ASME Design Engineering Division	2022
Awarded to 5 graduate students per year to uplift early career graduate students who are members of underrepresented groups within the design engineering division.	
Conference Participation Award, UConn	2021
Awarded to graduate students who actively participate in conferences, <i>Award amount: \$1000.</i>	
The 2nd Place, ME Graduate Research Competition, UConn	2021
Awarded by ME department at UConn, <i>Award amount: \$1500.</i>	
General Electric (GE) Fellowship, UConn	2020
Awarded to 10 graduate students with outstanding academic performance by the School of Engineering at UConn, <i>Award amount: \$15000.</i>	
The 2nd Prize of the National Mechanical Design Innovation Competition, China	2018
The National Excellence Prize of Casting Technique Competition, China	2018

Publications

• First-authored Journal Papers

- [J1] Wang, Z., Bhaduri, A., Xu, H., Wang, L., “An Uncertainty-aware Deep Learning Framework-based Robust Design Optimization of Metamaterial Units”. *Structural and Multidisciplinary Optimization* 68(3), 1–26 (2025).
- [J2] Wang, Z., Bray, A., Khanghah, K., Xu, H., “Designing Connectivity-Guaranteed Porous Metamaterial Units Using Generative Graph Neural Networks”. *Journal of Mechanical Design* 147, no. 2 (2025).
- [J3] Wang, Z., Xu, H., “Manufacturability-Aware Deep Generative Design of 3D Metamaterial Units for Additive Manufacturing”. *Structural and Multidisciplinary Optimization* 67, no. 2 (2024): 22.
- [J4] Wang, Z., Xian, W., Li, Y., Xu, H., “Embedding Physical Knowledge in Deep Neural Networks for Predicting the Phonon Dispersion Curves of Cellular Metamaterial”. *Computational Mechanics* 72, no. 1 (2023): 221-239.
- [J5] Wang, Z., Daeipour, M., Xu, H., “Quantification and Propagation of Aleatoric Uncertainties in Topological Structures”. *Reliability Engineering & System Safety* 233 (2023): 109122.
- [J6] Wang, Z., Xian, W., Baccouche, M., Lanzerath, H., Li, Y., Xu, H., “Design of Phononic Bandgap Metamaterials based on Gaussian Mixture Beta Variational Autoencoder and Iterative Model Updating”. *Journal of Mechanical Design*, (2022): 1-35.
- [J7] Wang, Z., Xu, H., “Quantitative Representation of Aleatoric Uncertainties in Network-like Topological Structural Systems”. *Journal of Mechanical Design*, (2021): 1-24.

• Co-authored Journal Papers

- [J8] Dolar, T., Ventura, N., Mignerot, F., Wang, Z., Kassner, C., Wadley, H., Howard, H., Gianola, D., Chen, W., “Accelerating Materials Discovery in Heterogeneous Composition-Property Design Spaces via Collaborative Bayesian Optimization”. *Materials & Design*, accepted.
- [J9] Kumbhojkar, G., Wang, Z., Keten, S., Chen, W., “Multi-objective Bayesian optimization for design under unknown feasibility constraints”. *Journal of Mechanical Design*, accepted.
- [J10] Cleeman, J., Jackson, A., Patel, A., Wang, Z., Feldhausen, T., Shao, C., Xu, H., Malhotra, R., “Operational Resilience of Additively Manufactured Parts to Stealthy Cyberphysical Attacks using Geometric and Process Digital Twins”. *Journal of Manufacturing Systems* 83 (2025): 626-651.
- [J11] Xu, L., Wang, Z. (Co-first), Liu, D., Rodgers, T., Tran, A., Xu, H., “Exascale Microstructure Reconstruction with Generative Deep Learning for CAE objects”. *Acta Materialia* 289 (2025): 120859.
- [J12] Song, J., Ihuaenyi, R., Lim, J., Wang, Z., Li, W., Fang, R., Ghamsari, A., Xu, H., Lee, Y., Zhu, J., “A Microstructural Electro-chemo-mechanical Model of High-nickel Composite Electrodes Towards Digital Twins to Bridge the Particle and Electrode-Level Characterizations”. *Energy & Environmental Science* 18, no. 7 (2025): 3129-3147.

- [J13] Efaw, C., **Wang, Z.**, Zhang, H., Xu, L., Kim, S., Park, B., Barnes, P., Xiong, C., Dufek, E., Khalifah, P., Xu, H., Li, B., "Pressure Effects on Lithium Anode/Nickel-Manganese-cobalt Oxide Cathode Pouch Cells Through Fixture Design". *Device* 3, no. 4 (2025).
- [J14] Kiarash, K., **Wang, Z.**, Xu, H., "Reconstruction and generation of porous metamaterial units via variational graph autoencoder and large language model". *Journal of Computing and Information Science in Engineering* 25, no. 2 (2025).
- [J15] Xu, L., Hoffman, N., **Wang, Z.**, Xu, H., "Harness Structural Stochasticity in Computational Discovery and Design of Microstructures". *Materials & Design* 223 (2022): 111223.
- [J16] Kheybari, M., **Wang, Z.**, Bilal, O., Xu, H., "Programmability of ultrathin metasurfaces through curvature". *Extreme Mechanics Letters* 52 (2022): 101620.

• **Journal Manuscripts Under Review**

- [J17] **Wang, Z.**, Dolar, T., Chen, Y., Chen, W., "ARCOBO: Adaptive Resource-aware Collaborative Bayesian Optimization for Heterogeneous Multi-Agent Design". Submitted to *Journal of Mechanical Design*, under review.

• **Refereed Conference Proceedings**

- [C1] **Wang, Z.**, Bray, A., Khanghah, K., Xu, H., "A Generative Graph Neural Network-based Framework for Designing Connectivity-Guaranteed Porous Metamaterial Units". in *International Design Engineering Technical Conferences & Computers and Information in Engineering Conference*, American Society of Mechanical Engineers, 2024.
- [C2] Kiarash, K., **Wang, Z.**, Xu, H., "Generating Porous Metamaterial Designs Using Variational Graph Autoencoder and Large Language Model". in *International Design Engineering Technical Conferences & Computers and Information in Engineering Conference*, American Society of Mechanical Engineers, 2024.
- [C3] **Wang, Z.**, Zhuang, R., Xian, W., Tian, J., Li, Y., Chen, S., Xu, H., "Phononic Metamaterial Design via Transfer Learning-based Topology Optimization Framework". in *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, American Society of Mechanical Engineers, 2022.
- [C4] **Wang, Z.**, Xian, W., Baccouche, R., Lanzerath, H., Li, Y., Xu, H., "A Gaussian Mixture Variational Autoencoder-Based Approach for Designing Phononic Bandgap Metamaterials". in *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, American Society of Mechanical Engineers, 2021.
- [C5] **Wang, Z.**, Xu, H., "Quantification of Uncertainties Distributed in Network-Like Systems". in *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, American Society of Mechanical Engineers, 2020.
- [C6] **Wang, Z.**, Xu, H., Li, Y., "Material Model Calibration by Deep Learning for Additively Manufactured Alloys". in *International Symposium on Flexible Automation*, American Society of Mechanical Engineers, 2020.

• **Invited Talks**

- [T1] "Material Image Analysis and Learning". ME5374 Scientific Machine Learning for Mechanical Engineers, Northeastern University. November 22, 2024.
- [T2] "Quantification of Aleatoric Uncertainties in Topological Structures & Physics-informed Machine Learning Model for Phononic Property Prediction". Probabilistic Seminar Series 2022, GE Research, US. September 28, 2022.

• **Conference Presentations & Posters**

- [P1] "Collaborative Bayesian Optimization under Heterogeneity: An Adaptive Resource-Aware Multi-Agent Design Framework." International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. CA. August 21, 2025.
- [P2] "A Generative Graph Neural Network-based Approach for Designing Connectivity-Guaranteed Porous Metamaterial Units." International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. DC. August 26, 2024.

- [P3] "Stochasticity Graded Structures Under Impact Loads." International Mechanical Engineering Congress & Exposition. Greater Columbus Convention Center, Columbus, OH. November 2, 2022.
- [P4] "Deep Learning-Assisted Cellular Metamaterial Design, Property Prediction and Uncertainty Analysis." B-Part Poster Presentation. International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. St. Louis. August 16, 2022.
- [P5] "Phononic Metamaterial Design via Transfer Learning-Based Topology Optimization Framework." International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. St. Louis. August 16, 2022.
- [P6] "Deep Generative Design of 3D Cellular Metamaterial." Shap3D IAB Meeting. Poster Presentation. UCONN, CT, May 5, Oct 4, 2022
- [P7] "Design of Phononic Bandgap Metamaterials with a Gaussian Mixture Variational Autoencoder-Based Iterative Search Approach." International Mechanical Engineering Congress & Exposition. Virtual, November 2, 2021.
- [P8] "A Gaussian Mixture Variational Autoencoder-Based Approach for Designing Phononic Bandgap Metamaterials." International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. Virtual. August 17, 2021.
- [P9] "Quantification of Uncertainties Distributed in Network-Like Systems." International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. Virtual. August 18, 2020.

Research Experience

Postdoctoral Scholar, Department of Mechanical Engineering, Northwestern University Sep 2024 – Present

Project: (AFOSR MURI) Compositionally Complex Ceramics for Hypersonics via Knowledge-Guided Ceramization.

- Graph neural network-based machine learning for optimization of ReaxFF parameterization.
Collaborator: (PSU) Prof. Adri van Duin
- Mixed-variable adaptive material and process co-design via Latent Variable Gaussian Process (LVGP) modeling and Bayesian Optimization.
Collaborator: (ASU) Prof. Timothy Long

Project: (ARO HTMAX) Center on High-Throughput Materials Discovery for Extremes.

- AI-based continual learning methods for autonomous high-throughput X-ray diffraction image classification
Collaborator: (JHU) Professor Todd Hufnagel
- Collaborative Bayesian optimization for accelerating materials discovery in heterogeneous design spaces.
Collaborator: (UCSB) Professor Daniel S. Gianola

Graduate Research Assistant, Mechanical Engineering, UConn Sep 2019 – Aug 2024

Project: (NSF CAREER) Deep learning-driven design of metamaterial considering manufacturing uncertainties.

- Deep generative discovery and design of metamaterials to achieve superior performances (e.g., elasticity, thermal behavior, wave trapping) while considering manufacturability constraints.
- Robust design of metamaterials using probabilistic deep generative model considering both aleatoric uncertainty and epistemic uncertainty.

Project: (FORD URP) Physics-guided deep learning for design performance evaluation.

- Physics-embedded deep learning for metamaterial performance evaluation with limited amount of data.
Collaborator: (UW-Madison) Professor Ying Li
- Machine learning-based model calibration for predicting the unknown material parameters in additively manufactured alloys.

Project: Uncertainty quantification, uncertainty propagation, and design for reliability/robustness.

- Uncertainty quantification and propagation of aleatoric uncertainties in complex topological structures.

- Computational design of lattice structures for vehicle impact performance with structural uncertainties.

Undergraduate Research Assistant, HUST, Wuhan, China

Sep 2024 – Present

- Design and construction of a double-lifting parking device, which offers triple parking capacity.
- Mechanical analysis and numerical simulation using ABAQUS to validate structural integrity.

Proposal Experience

NSF PCL

2025

Test Bed: Toward a Network of Programmable Cloud Laboratories.

NSF DMREF

2025

Interface Design-enabled Microstructure Architecture in Non-woven Fabric Materials towards Tunable Mechanical Strengths and Biodegradability.

DOE BES

2025

Multi-scale Integration of Experiments, Modeling, and Machine Learning to Unravel Microstructure Development in Laser-Induced Alloying of Compositionally Complex Alloy.

Industry Experience

Fellow Intern

May 2023 – Aug 2023

General Electric (GE) Global Research, Niskayuna, NY, USA

- Collaborated with GE business design and services communities in the development of methods for probabilistic design, machine learning and optimization.
- Implemented probabilistic design, machine learning and optimization methods into GE internal design.

Teaching Experience

Teaching Apprentice, Northwestern University

Winter 2025

- Assisting in teaching ME 441 by setting up homework assignments, exams, and supervising student projects.
- Delivering 1–2 lectures and attending regular class sessions.
- Working closely with the lead instructor to manage course structure and pedagogy.

Graduate Teaching Assistant, UConn

Aug 2019 - May 2022

- Worked as teaching assistant in three undergraduate-level courses and one graduate-level course: MEM3221: Introduction to Products & Process, ME3221: Manufacturing Automation, ME3227: Design of Machine Elements, ME/SE5702: Data Science for Materials and Manufacturing.
- Held office hours, graded homework and exams, provided individual/group tutoring.

DaVinci Teaching Assistant, UConn

July 2021, July 2022

- Worked as a Teaching Assistant in the UCONN DaVinci Project, helped K–12 teachers enrich classroom activities by incorporating engineering into the curriculum.
- Taught engineering fundamentals to K-12 teachers and assisted them in developing practical curricula and exercises to introduce students to engineering.

Mentorship Experience

Bolin Chen, (Ph.D. program, Northwestern)

2025 - date

Mentoring on developing LLM for material design and discovery.

Christian Fernandez Soria, (Ph.D. program, Northwestern)

2025 - date

Mentoring on Bayesian optimization algorithms and their implementation in materials discovery tasks.

Gourav Kumbhojkar, (Ph.D. program, Northwestern)

2024 - date

Mentored on developing a latent variable Gaussian process model for classification.

Output: One joint journal paper [J9]

Tuba Dolar , (Ph.D. program, Northwestern)	2024 - 2025
Mentoring on developing collaborative Bayesian doptimization algorithm for material discovery.	
<u>Output</u> : One joint journal paper [J8]	
Austin Bray , (MS program, UConn)	2022 - 2024
Tutored on developing deep learning model for fluid-infilled metamaterials. Mentored the development of simulation model for creating datasets.	
<u>Output</u> : One journal paper [J2], one conference paper [C1].	
Filip Penda , (MS program, UConn)	2022 - 2024
Mentored on developing Abaqus simulation model for modeling fracture and battery mechanics analysis.	
Nathon Hoffman , (MS program, UConn)	2020 - 2022
Mentored on developing simulation models for modeling impact and crash.	
<u>Output</u> : One joint journal paper [J15].	

Professional Services and Membership

Invited reviewer of Additive Manufacturing (impact factor: 11.1)

Invited reviewer of International Journal of Mechanical Sciences (impact factor: 9.4)

Invited reviewer of npj metamaterials (impact factor: 8.3)

Invited reviewer of Engineering Applications of Artificial Intelligence (impact factor: 8.0)

Invited reviewer of Engineering Structures (impact factor: 8.0)

Invited reviewer of Knowledge-Based Systems (impact factor: 8.0)

Invited reviewer of Measurement (impact factor: 5.6)

Invited reviewer of Progress in Additive Manufacturing (impact factor: 5.4)

Invited reviewer of Applied Mathematical Modelling (impact factor: 5.1)

Invited reviewer of Structural and Multidisciplinary Optimization (impact factor: 4.279)

Invited reviewer of Scientific Reports (impact factor: 3.8)

Invited reviewer of Journal of Mechanical Design (impact factor: 3.441)

Invited reviewer of International Journal for Numerical Methods in Engineering (impact factor: 2.9)

Invited reviewer of Plasmonics (impact factor: 2.726)

Invited reviewer of ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems

Invited reviewer of SAE International Journal of Materials and Manufacturing

Invited reviewer of the ASME IDETC conferences

ASME Student Member